

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8872/01

Wednesday 29 August 2012
50 minutes

Additional Materials: Multiple Choice Answer Booklet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

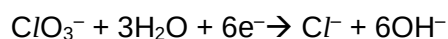
Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

- 1** NaN₃ was commonly used in early designs of airbags for cars. When a crash occurs, an electronic trigger will result in the decomposition of NaN₃ to give sodium metal and nitrogen gas as the only products. A typical airbag can provide adequate protection for the driver during a car crash, with full inflation that requires around 28 dm³ of nitrogen. What would be the minimum mass of NaN₃ that can produce the required gas?

A 152.1g **B** 114.1g **C** 76.1g **D** 50.7g

- 2** Chlorate (V) ions, ClO₃⁻ on heating will disproportionate to chlorate (VII) ions, ClO₄⁻ ions and chloride ions in the following half-equations.



Given that only 0.6 mol of ClO₄⁻ ions is formed, what is the number of moles of chloride ions formed?

A 0.1 **B** 0.2 **C** 0.3 **D** 0.6

- 3** For complete combustion of 10 cm³ of a gaseous hydrocarbon compound, 30 cm³ of oxygen gas was required.

What could be the formula of the compound?

A CH₂CH₂ **B** C₂H₂ **C** CH₃CH₃ **D** CH₃CHO

- 4** Which of the following elements contains two unpaired electrons?

A Nitrogen **B** Oxygen
C Neon **D** Magnesium

- 5 Isotopes have many uses in research for understanding of chemistry and two often used isotopes are $^{14}_6\text{C}$ and $^{18}_8\text{O}$.

Which of the following statements regarding these isotopes is correct?

- A** Both the carbon and oxygen isotopes have the same number of neutrons.
- B** Both the carbon and oxygen isotopes have the same number of protons.
- C** The oxygen isotope has the same number of protons as the number of neutrons in the carbon isotope.
- D** The oxygen isotope can be converted to the carbon isotope with the removal of 4 protons from the nucleus.
- 6 Blu-Tack is a reusable adhesive that consists of a synthetic long chain hydrocarbon, and it is able to stick on to many surfaces, such as ceramic tiles and glass.

What is the likely attraction between Blu-Tack and glass surfaces?

- A** van der Waal's forces **B** covalent bond
- C** covalent dative bond **D** ionic bonds
- 7 Which of the following species is **not** planar?
- A** Benzene, C_6H_6 **B** Aluminium Chloride, AlCl_3
- C** Methanal, CH_2O **D** Sulfate (IV), SO_3^{2-}

- 8 Which set of the following properties would typically apply to a metal?

	Melting point ($^{\circ}\text{C}$)	Solubility in water	Electrical conductivity of the molten compound
A	110	insoluble	does not conduct
B	808	very soluble	conducts well
C	1610	insoluble	does not conduct
D	1085	insoluble	conducts well

- 9 Which of the following ionic compounds is expected to have the highest numerical value for its lattice energy?

A Rb₂O B RbF C SrO D SrF₂

- 10 Dinitrogen pentoxide, N₂O₅ can be produced from the reaction of NO₂ and O₂. 1 mol of N₂O₅ formed has the enthalpy change of $\Delta H = x$.

Reactions	$\Delta H / \text{kJ mol}^{-1}$
$2\text{NO (g)} + \frac{3}{2}\text{O}_2 \text{ (g)} \rightarrow \text{N}_2\text{O}_5 \text{ (g)}$	-169
$\text{NO (g)} + \frac{1}{2}\text{O}_2 \text{ (g)} \rightarrow \text{NO}_2 \text{ (g)}$	-57

By using the data above, what is the correct value for x in kJ mol^{-1} ?

A +112 B +55 C -55 D -112

- 11 Dinitrogen tetraoxide dissociates at a temperature of 60 °C according to the following equation.



Which of the following changes, when applied to a system of N₂O₄ and NO₂ at equilibrium, will result in the corresponding outcome?

	Change to the system	Position of Equilibrium	Equilibrium constant, K_c
A	adding more N ₂ O ₄	shift right	increases
B	increase pressure	shift left	no change
C	increase temperature	shift right	no change
D	adding catalyst	No change	increases

- 12 2 mol of hypoiodous acid, HOI was dissolved in 1 dm³ of water and produced a solution of pH 5.2.

Given the above information, which of the following statement is true?

- A HOI is a weak acid with a small K_a value.
 B HOI is a weak acid with a large K_a value.
 C HOI is a strong acid which dissociates fully in the solution.
 D The concentration of H⁺ in the above solution is about 0.0055 mol dm⁻³.

- 13** Many food and drinks produced commercially may use phosphoric acid and its salts as additives to act as a buffer.

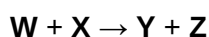
Which statement about buffer solutions is correct?

- A** The pH of a buffer solution remains absolutely unchanged when small amounts of acid or base are added.
 - B** The pH of a buffer solution decreases very slightly when small amounts of acid are added.
 - C** The pH of a buffer solution decreases very slightly when small amounts of base are added.
 - D** The pH of a buffer solution changes slightly when very large amounts of acid or base are added.
- 14** At 25°C, hydrogen peroxide solution decomposes according to the rate equation below.

$$\text{Rate} = k[\text{H}_2\text{O}_2]$$

Given that the rate constant, k , is 6.33×10^{-3} , how long does it take for the concentration of hydrogen peroxide to reduce to 25% of the original sample in the reaction flask?

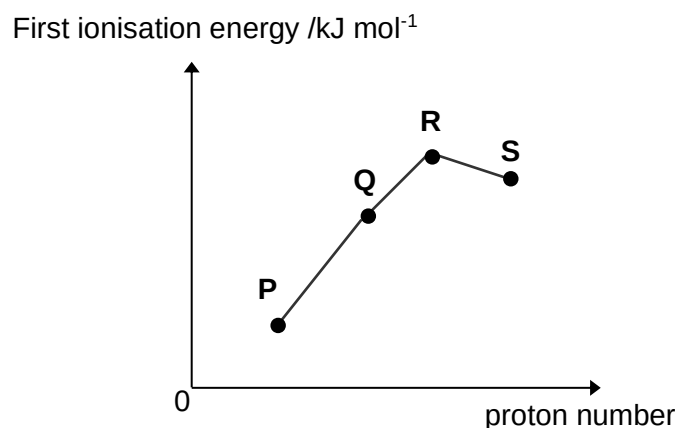
- A** 110s **B** 219s **C** 329s **D** 438s
- 15** If the reaction,



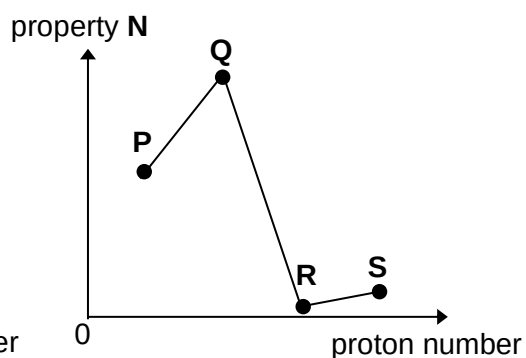
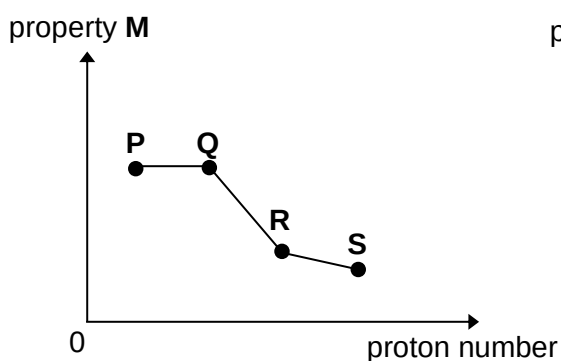
is described as being first order with respect to **W**, which of the following statements must be true?

- A** **X** is acting as a catalyst.
- B** The half-life of the reaction is constant.
- C** All molecules of **W** possess sufficient activation energy to react.
- D** The rate of reaction is directly proportional to the concentration of **W**.

- 16** The diagram represents the first ionisation energy of four consecutive elements in the third period of the Periodic Table, with element **R** showing two oxidation states in its chlorides.



The sketches below represent another two properties of these elements.



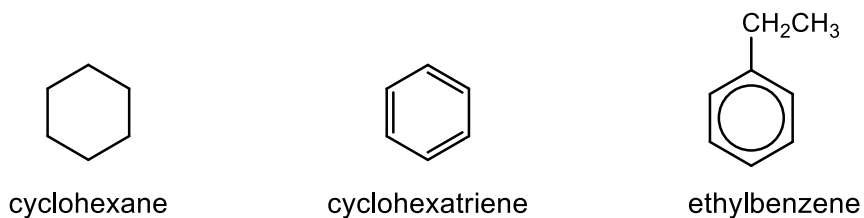
What are properties **M** and **N**?

property M	property N
A pH of oxides in water	boiling point
B ionic charge	4 th ionisation energy
C ionic radius	nuclear charge
D electrical conductivity	atomic radius

- 17 Element **H** is found in Period 3 of the periodic table. Oxide of element **H** dissolves sparingly in water to produce a solution that turns red litmus paper blue. However, chloride of element **H** dissolved in water to produce a yellowish-green solution when universal indicator is added.

What could be the identity of element **H**?

- A Na B Mg C Al D P
- 18 How many geometric isomers, can be formed from a compound of molecular formula $C_3H_4Cl_2$?
- A 2 B 3 C 4 D 5
- 19 An experiment was carried out and the following compounds were reacted **in the same reaction flask** with excess hot concentrated alkaline potassium manganate(VII) solution.



Which of the following will be observed?

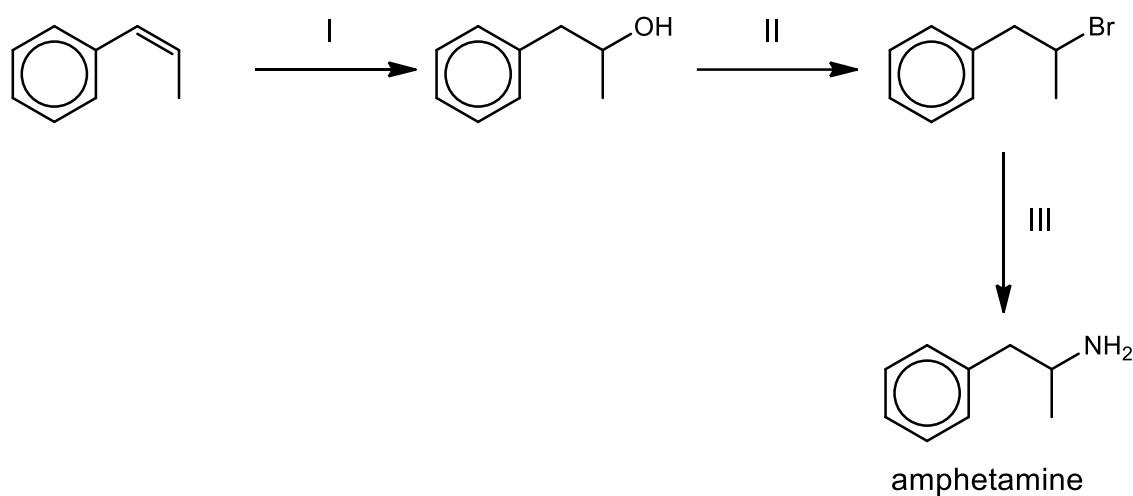
- A No effervescence will be observed.
- B A colourless solution is obtained
- C The resultant mixture obtained turned blue litmus paper red.
- D The resultant mixture has an aqueous layer and an organic layer.

- 20 Compound **T** is reacted with excess NaOH(aq) while compound **V** is reacted with LiAlH₄ in dry ether.

Which of the following pairs of **T** and **V** give the same product?

	T	V
A	BrCH ₂ CH ₂ CH ₂ OH	CHOCH ₂ CO ₂ H
B	CH ₃ CH ₂ BrCH ₃	CH ₃ CO ₂ CH ₃
C	CH ₃ CH ₂ CH ₂ Cl	(CH ₃) ₂ CO
D	C ₆ H ₅ Br	C ₆ H ₅ CO ₂ H

- 21 Amphetamine, is a psychostimulant drug that induces increased wakefulness and focus in association with decreased fatigue and appetite. It can be produced via a two-step reaction as shown below.



What are the reagents and conditions required at step I and step II?

	I	II	III
A	conc. H ₂ SO ₄ , reflux	PBr ₃ , reflux	conc. NH ₃ (aq), heated in sealed tube
B	1. cold H ₂ SO ₄ 2. water, heat	NaBr, conc. H ₂ SO ₄ , reflux	conc. NH ₃ (alc), heated in sealed tube
C	steam, conc. H ₃ PO ₄ , 70 atm, 300 °C	conc. HBr	conc. NH ₃ (aq), heated in sealed tube
D	dil. H ₂ SO ₄ (aq), reflux	PBr ₃ , rtp	conc. NH ₃ (alc), heated in sealed tube

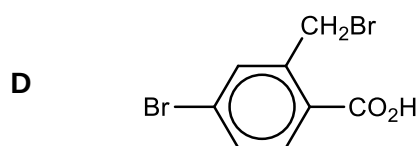
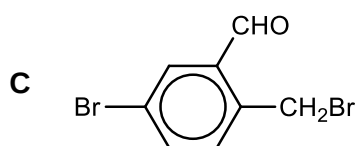
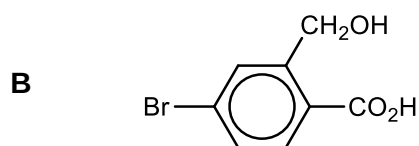
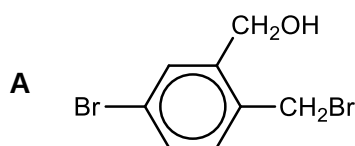
22 Which sequence shows the compounds in order of decreasing pK_a values?

- A $\text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CO}_2\text{H} > \text{CH}_2\text{FCO}_2\text{H} > \text{CH}_2\text{ClCO}_2\text{H}$
 B $\text{CH}_2\text{FCO}_2\text{H} > \text{CH}_3\text{CO}_2\text{H} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_2\text{ClCO}_2\text{H}$
 C $\text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CO}_2\text{H} > \text{CH}_2\text{ClCO}_2\text{H} > \text{CH}_2\text{FCO}_2\text{H}$
 D $\text{CH}_2\text{ClCO}_2\text{H} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CO}_2\text{H} > \text{CH}_2\text{FCO}_2\text{H}$

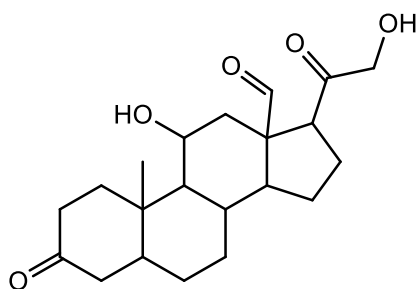
23 An organic compound **K** has the following properties:

- 1 mole of **K** reacts with excess Na(s) to produce one mole of $\text{H}_2(\text{g})$
- 1 mole of **K** reacts with only one mole of conc. HCl .

Which compound could **K** be?



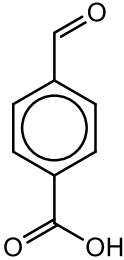
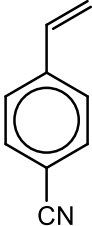
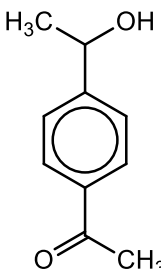
24 Aldosterone is a steroid hormone that increases reabsorption of ions and water in the kidney. It has the structure as shown below.



Which of the following regarding aldosterone is **false**?

- A It reacts with 2,4-dinitrophenylhydrazine to produce orange precipitate.
 B 2 mole of $\text{H}_2(\text{g})$ is produced when reacting with Na(s) .
 C It reacts with 2 moles of PCl_5 to produce HCl .
 D It reacts with Fehling's reagent to give brick red precipitate.

25 Which of the following reactions would **not** produce 1,4-dicarboxylic acid?

	Reactant	Reagents and conditions
A		hot acidified potassium dichromate(VI)
B		hot alkaline potassium manganate (VII)
C		dilute H ₂ SO ₄ , heat
D		alkaline aqueous iodine and heat followed by acidification

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

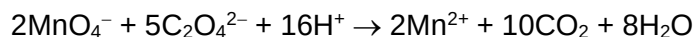
No other combination of statements is used as a correct response.

- 26** Diamonds are often sold as valuable gemstones due to their appearance which allows light to pass through. Diamonds are also used in industry in cutting and grinding tools due to the hardness of the element. Graphite on the other hand has a dull appearance and has applications as lubricants and electrical conductors.

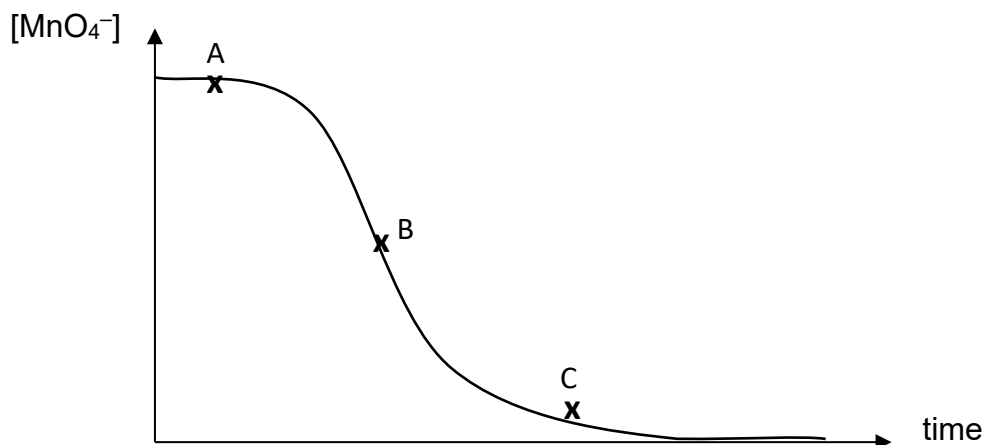
Which of the following statements is correct in the explanation of some physical properties of diamond and graphite?

- 1** Graphite has delocalised electrons between layers which conduct electricity along the layer.
- 2** Diamond has an open giant covalent structure, which allows light to pass through, and this rigid tetrahedral structure also results in its hardness.
- 3** Graphite has weak van der Waal's forces of attraction between layers, hence the layers can slide past each other easily which results in its lubricating property.

- 27 The reaction of permanganate(VII), MnO_4^- , with oxalate, $\text{C}_2\text{O}_4^{2-}$ is a type of autocatalytic reaction whereby the product, manganate(II) ions (Mn^{2+}), is the catalyst for the reaction.



The following graph was obtained when the rate of the above reaction was plotted against time.



Which of the following can be deduced from the graph?

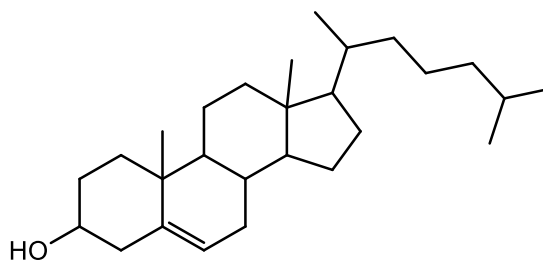
- 1 Rate at point A is very slow as two anions are reacted together.
 - 2 Rate at point B has increased due to the product of Mn^{2+} ions.
 - 3 Rate at point C is slow due to the decrease in [reactants]
- 28 The bond length of Group VII elements are shown below.

element X	Cl_2	Br_2	I_2
bond length X – X / nm	0.199	0.228	0.266

Which of the following explains the increase in bond length down the group?

- 1 Atomic radius increases down the group.
- 2 Orbital overlap decreases down the group.
- 3 Nuclear charge increases down the group.

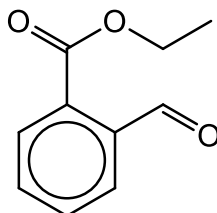
- 29** The diagram shows the structure of the naturally-occurring molecule, cholesterol.



cholesterol

Which statements about cholesterol are correct?

- 1** All carbons in the molecule lie on the same plane.
 - 2** The molecule does not exhibit any geometrical isomerism.
 - 3** Substitution can occur in the presence of chlorine gas and UV light.
- 30** Compound **Y** has the following structure.



Compound **Y**

Which of the following statements are true?

- 1** Red ppt is observed when compound **Y** is heated with Fehling's reagent.
- 2** Yellow ppt is observed when compound **Y** is heated with alkaline aqueous iodine solution.
- 3** White ppt is observed when compound **Y** is heated with acidified potassium manganate(VII).

~ END OF PAPER ~